

Ellan Vannin Wealth Management

Sovereign Infrastructure Investment Memorandum

The Humber Bitcoin Energy Reserve Programme

A Strategic Bitcoin Mining & Energy Arbitrage Investment Strategy (2026–2040)

Turning Yorkshire's Future Energy Surplus into a Digital Sovereign Asset

Executive Summary

Ellan Vannin Wealth Management (EVWM) proposes the creation of:

The Humber Digital Energy Fund

A long-term infrastructure investment vehicle combining:

- offshore wind power
- Humber industrial regeneration
- battery storage
- flexible electricity demand
- Bitcoin mining infrastructure

The investment thesis:

When Britain produces large amounts of cheap renewable electricity, the ability to convert excess energy into a globally liquid monetary asset becomes a strategic industrial advantage.

Hull is uniquely positioned for such a strategy because it is already developing as a renewable-energy manufacturing hub, with Green Port Hull supporting offshore wind supply chains and renewable-energy investment.

The wider Humber region is also positioned as a clean-energy industrial cluster through offshore wind, hydrogen and industrial decarbonisation projects.

Investment Scenario

Assumptions

This is a hypothetical Anglo-futurist infrastructure scenario.

Bitcoin price:

£1,000,000 per BTC by 2040

Investment period:

2026–2040

14 years

Initial capital deployment:

£250 million

Allocation:

Asset	Investment
Mining facility	£100m
ASIC hardware	£75m
Energy infrastructure	£40m
Grid connection/storage	£25m
Operations	£10m

The Strategic Advantage: Cheap Humber Energy

The thesis assumes the Humber becomes a major energy hub through:

- offshore wind
- tidal projects
- hydrogen infrastructure
- grid upgrades
- industrial electrification

Hull already sits within Britain's offshore wind supply-chain ecosystem, including major turbine manufacturing and port infrastructure.

Proposed Facility

Humber Bitcoin Energy Campus

Location:

Hull / East Yorkshire industrial zone

Capacity:

100 MW Bitcoin mining operation

Electricity requirement:

100 MW continuous load

Annual consumption:

≈876 GWh

Electricity cost assumption:

Industrial renewable power:

£30/MWh

Annual electricity cost:

£26.3 million

Mining Economics

Bitcoin mining economics depend on:

- BTC price
- network difficulty
- hardware efficiency
- electricity costs

Mining profitability becomes increasingly competitive as global hash rate rises, so this model assumes technological improvement and efficient energy access rather than today's economics.

Bitcoin mining difficulty and hardware requirements have historically increased over time as miners compete for block rewards.

Mining Output Assumption

2026

100 MW facility

Produces:

≈450 BTC/year

Assuming efficiency improvements but increased network difficulty:

Year BTC mined

2026 450

2030 350

2035 250

2040 150

Total mined:

~4,500 BTC

Bitcoin Value Scenario

Assumption:

£1,000,000 BTC in 2040

Portfolio BTC value:

4,500 BTC × £1m

=

£4.5 billion

Revenue Model

Annual BTC Production Value

2026

BTC price assumption:

£100,000

Mining:

450 BTC

Revenue:

£45m

Electricity:

£26m

Operating costs:

£8m

Operating cash flow:

£11m

2030 Scenario

BTC price:

£250,000

Mining:

350 BTC

Revenue:

£87.5m

Costs:

£40m

Operating profit:

£47.5m

2035 Scenario

BTC price:

£500,000

Mining:

250 BTC

Revenue:

£125m

Costs:

£50m

Operating profit:

£75m

2040 Scenario

BTC price:

£1,000,000

Mining:

150 BTC

Revenue:

£150m

Costs:

£60m

Operating profit:

£90m

14-Year Financial Model

Total BTC mined:

4,500 BTC

Total electricity expenditure:

≈£600m

Total operational expenditure:

≈£150m

Total BTC asset value:

£4.5bn

Internal Rate of Return (IRR)

Initial investment:

£250m

Terminal asset value:

£4.5bn

Ignoring interim cash flows:

IRR:

≈22%

Including operating profits:

Estimated IRR:

25–30%

Sensitivity Analysis

Bear Case

BTC reaches:

£250,000

BTC holdings:

4,500

Asset value:

£1.125bn

IRR:

≈12–15%

Base Case

BTC:

£1m

Asset value:

£4.5bn

IRR:

≈25%

Bull Case

BTC:

£2m

Asset value:

£9bn

IRR:

≈35%+

Why Hull?

The investment is not purely about Bitcoin.

It is about energy monetisation.

A future Humber economy could combine:

Offshore Wind

↓

cheap electricity

↓

Bitcoin Mining

↓

monetise surplus electricity

↓

Digital Asset Reserve

↓

finance further infrastructure

Similar concepts are emerging internationally where miners locate near abundant energy sources and act as flexible electricity consumers. Bitcoin mining economics are strongly tied to electricity prices and hardware efficiency.

Wider Economic Impact

Direct Employment

A 100 MW automated facility:

50–100 permanent workers

But the larger effect comes from:

- electrical engineering
 - data-centre construction
 - grid infrastructure
 - renewable energy development
 - cybersecurity
 - AI/data infrastructure
-

Strategic Option: Bitcoin + AI Campus

By 2040 the facility could become:

Humber Digital Energy Campus

Containing:

- Bitcoin mining

- AI compute
- battery storage
- industrial heat recovery
- grid balancing

The same power infrastructure supports multiple future industries.

Integration with EVWM Strategy

The Humber Bitcoin Energy Reserve connects:

Port Expansion

↓

Offshore Energy

↓

Cheap Electricity

↓

Digital Infrastructure

↓

Sovereign Asset Accumulation

Final Recommendation

Ellan Vannin Wealth Management recommends establishing:

The Humber Digital Energy Fund

A £250 million infrastructure investment programme to create a strategic energy-to-digital asset facility in Hull.

The investment thesis:

Britain historically converted coal into industrial power, oil into transport wealth, and finance into global influence. An Anglo-futurist economy could convert renewable electricity into digital capital.

Under the scenario of:

£1 million Bitcoin by 2040

a well-capitalised Humber mining operation could theoretically generate:

- multi-billion-pound asset exposure
- 20%+ infrastructure-style returns
- regional technology investment
- a new category of energy monetisation

The broader idea is not simply "Bitcoin mining".

It is:

turning Britain's future energy abundance into a sovereign digital reserve asset.